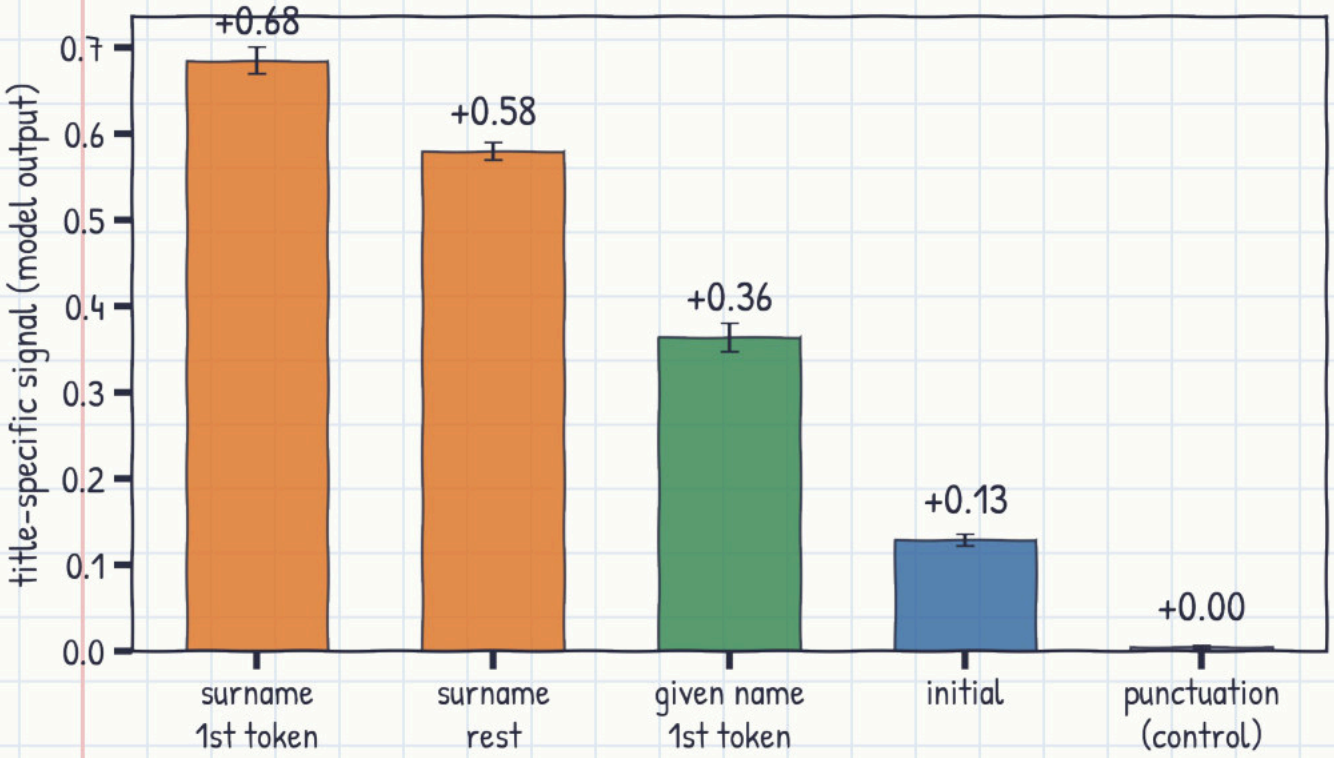
Where the signal sits, and a dose-response

model output

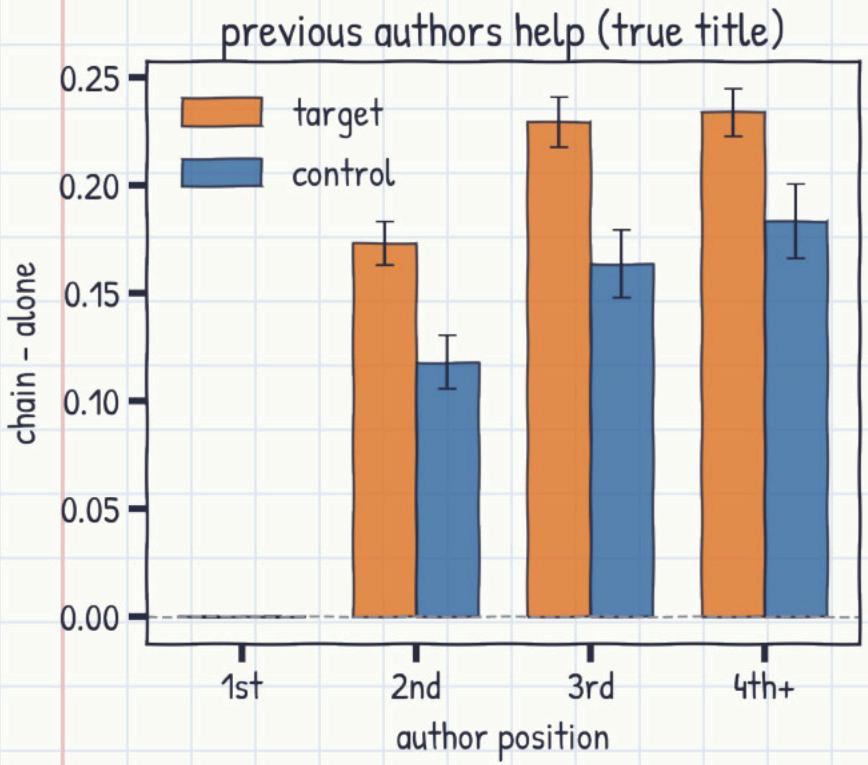


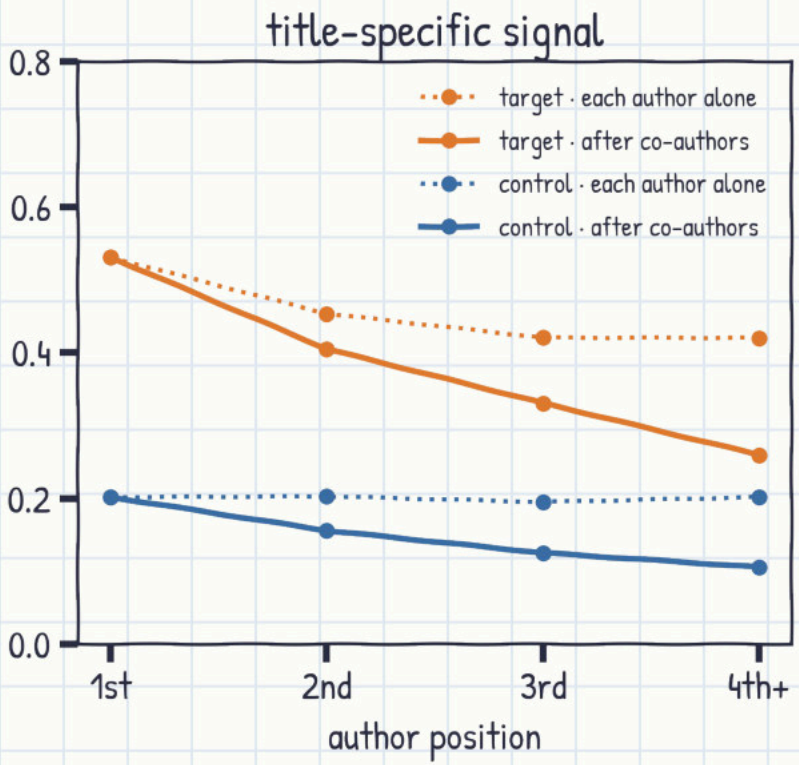


Right: R0/R1 papers, centred within citation bin; Spearman rho = 0.27, $p < 0.001$. (Stage 10, 8,894 papers, §6.6)

Co-authors help, and partly replace the title

model output · all 9,108 papers · 95% CI (Stage 11 E3, §6.11)





later authors **+0.17 to +0.23** (Stage 11 E3, 4,621 target, §6.11)

title-specific part **shrinks**

target > control still

Stages 1–9 · 20–200 papers

What we saw

- A–B lives on continuation tokens (+0.021), not the first token (+0.001)(7.1, 20 pairs)
- Open question: is that teacher forcing, initials or surnames?

Stages 10–11 · 9,108 papers

What scale showed

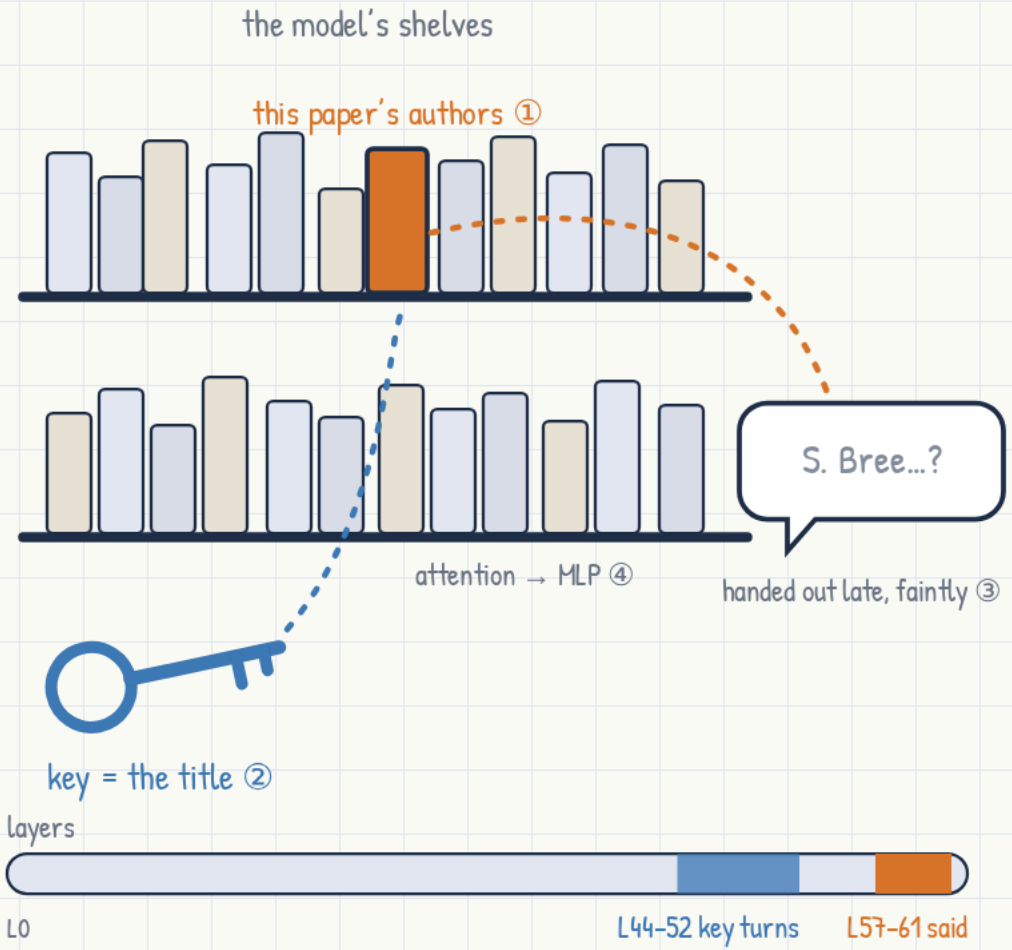
- 73% of names start with an initial(24,739 / 33,780; §6.7), which carries little
- Full given names carry as much as surnames and help predict them (+0.22)(final L63; §6.7)
- Information travels title → name positions (L44–52), partly via tail tokens
- Earlier co-authors make later ones easier, title matters less

Line D: the puzzle from 7.1 is explained by name format

The pilot's "continuation" result was mostly a format effect: for "S. Bree" the surname is a continuation token. Stage 12 P1 (done) patched single positions: the name positions carry the signal (+0.85 at L52), the colon alone little (+0.17)(P1, 1,186 papers; P1_patch_pos.csv).

Keys and shelves

Is the book on the shelf, or is the key failing to get it out?



怎么算：spec = true − mean(false_0..4)；recovery(l) = mean_p[P̄_p(l) − F̄_p] / mean_p[T̄_p − F̄_p]；T − C 档内合并 · 9,108 篇（lens）/ 1,186 篇（E1、P1、P2，P1/P2 为 5 个假标题平均）· 出处：X3_means.csv、E1_recovery.csv、p12f_mean5.csv

1 Is the book on the shelf? Yes

True title beats false titles (≈0.44 at L62(§6.9)); a probe on the title token reads paper-specific authors from L24 (re-checked)

Lines A, B · Stage 5, 7.3, 8D, 10, 11

2 Key = the title, read into the name positions at L44–52

Whole prompt: 0.82 at L48, ≈1.0 from L52; name positions alone 0.613 → 0.841 (false_0 0.61 → 0.85); ':' ≤ 0.181 (0.17)(E1, P1 5-title mean; 1,186 papers); knockout after L52 costs almost nothing

Line C · Stage 11, 12 P1

3 Book there but stuck? Not supported

Target − control ≈ 0 at L24–L40, slightly negative at L48 (−0.04 to −0.05), grows only at L57–61 (+0.21 at output(§6.4)): retrieval itself is weak

Lines A, C · Stage 7.2, 10

4 Which shelf? Both, in order

Attention brings the title in at L48 (0.237; false_0 0.231); MLP takes over at L52–56 (0.181, 0.226; false_0 0.178, 0.224)(P2 5-title mean, p12f_mean5.csv). One-layer patching is a lower bound; whether the MLP step is memory lookup is not yet separable

Stage 12 P2

Caveat: a readout not finding it ≠ absent. Stage 13: ':' and name positions hold the task; the title position holds paper attributes and meaning beyond the words, not shown to be paper identity.

What the small samples taught us to watch for

- **Single-paper leverage**: one paper = 71% of the Stage 4 L56 mean (20 papers)
- **Tail-driven contrasts**: Stage 8 A–B disappears after removing 10 of 100 pairs (8D, matched_pair_differences.csv)
- **Forking paths**: L56 inspected repeatedly, then made primary
- **Candidate artefacts**: 100 false names share ~46 first tokens; 3% tie with gold (7.2 quality.json: 156/5,000)
- **Label noise**: strict "pass" 25.5% in the 0–2 citation bin (label_README.md)
- **Partial coverage**: first-listed author stands in for the team
- **Metric choice**: KL fidelity and author ranking disagree
- **Protocol drift**: plan locked 10-18 14:29 (7-layer integral primary); the 21:14 PROTOCOL switched to L56, both before 8C scoring; who and why open (U28)

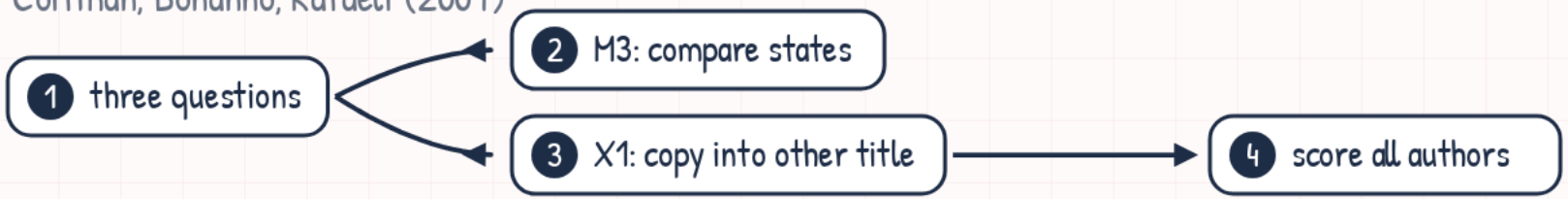
Stage 10 answered each one by design: full cohort, every author and token, all layers reported, full-vocabulary ranks, within-bin comparison, frozen inputs with parity checks. What it cannot fix: groups still come from one labelling run of the same model.

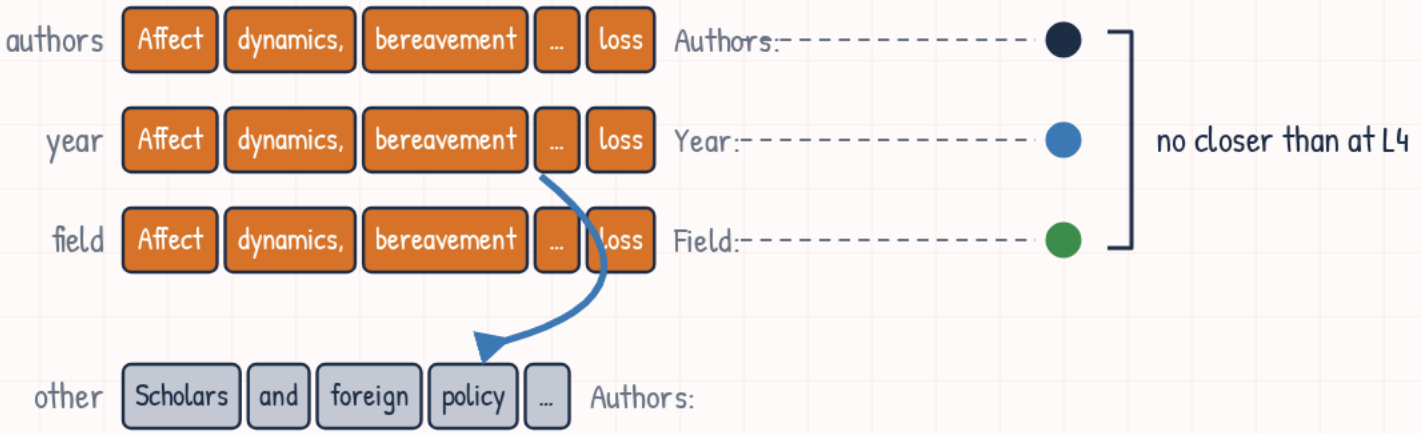
Small vs large, in one table

Finding	Stages 1–9 (20–200 papers)	Stages 10–11 (9,108 papers)	Verdict
True title helps	+9.3 pp pilot, +5.2 pp new 200 (7.3; 8D RQ1)	≈0.44 title-specific at L62 (X3_means.csv)	confirmed
Recognised > not recognised	uncertain, tail-driven	+0.21 at output (§6.4), ≈0 at L24–L40, slightly negative at L48 (−0.04 to −0.05)	confirmed, late only
Same groups (A vs B, R0 only)	A − B as above	+0.20 at output (§6.13), ≈0 at L24–L40, slightly negative at L48; R1 adds strength, not timing	comparable, unchanged
Key layer	L56	starts L56, rises L57–61; read L24–50	refined
J-Lens over Vanilla	no increment	more name shape, no paper identity	consistent
First vs rest of name	gap on continuation	explained by initials vs given names	explained
Citations	low–high inconclusive	monotonic, ρ = 0.451 (X2_spearman.csv)	resolved
Lens stability	split-half one split; Stage 9 only Fit A	three Jacobian lenses fitted on different text agree; fit stability rests on Stage 9 Fit A only	only one piece of evidence (Stage 9 Fit A); Fit B not completed

Changing the question

Coifman, Bonanno, Rafaeli (2007)





R from	authors	year	field
L24	**0.996**	**0.990**	**0.992**
L40	**0.933**	**0.928**	**0.929**
L48	**0.455**	**0.431**	**0.444**

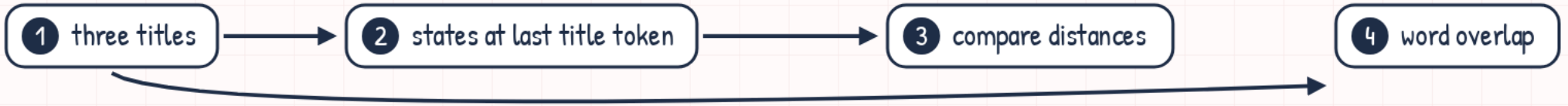
R(l) = (patched − other) ÷ (own − other)

title states copied at layer l; 582 test papers

R: Stage 13 X1, all authors (batch 1 item 1), test 582 papers; b1_xall_report.md (old first-author: 0.99 / 0.92 / 0.42)

Paraphrased title (X3)

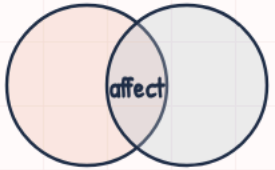
same paper in other words vs other paper, same words







para 0.22



D 0.14

G′ = cos(orig, para) − cos(orig, D)

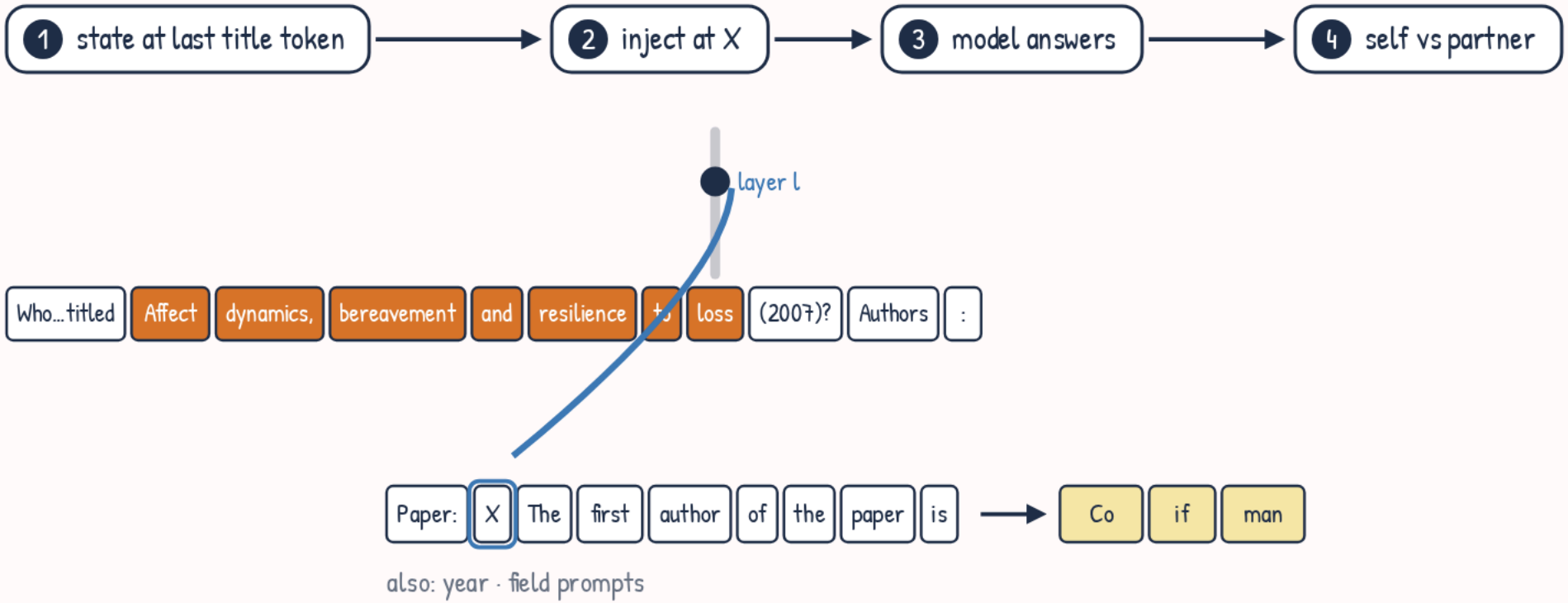
+0.32 at L8 · peak +0.46 at L36 (test set)

medians: 0.61 vs 0.15

all-author score, paraphrase − D: +0.229

Patchscope (X4)

Coifman, Bonanno, Rafaeli (2007)



D = score(own state) − score(partner state)

score = log10(1/rank) of the true answer tokens

first author from L8: +0.046

year from L40: +0.013

D: Stage 13 X4 (prompt asks for the first author by design), test set, team CI; first author L8 +0.046 [0.035, 0.059], year L40 +0.013 [0.008, 0.018]; x4_report.md

Stage 12–13: what the middle layers hold

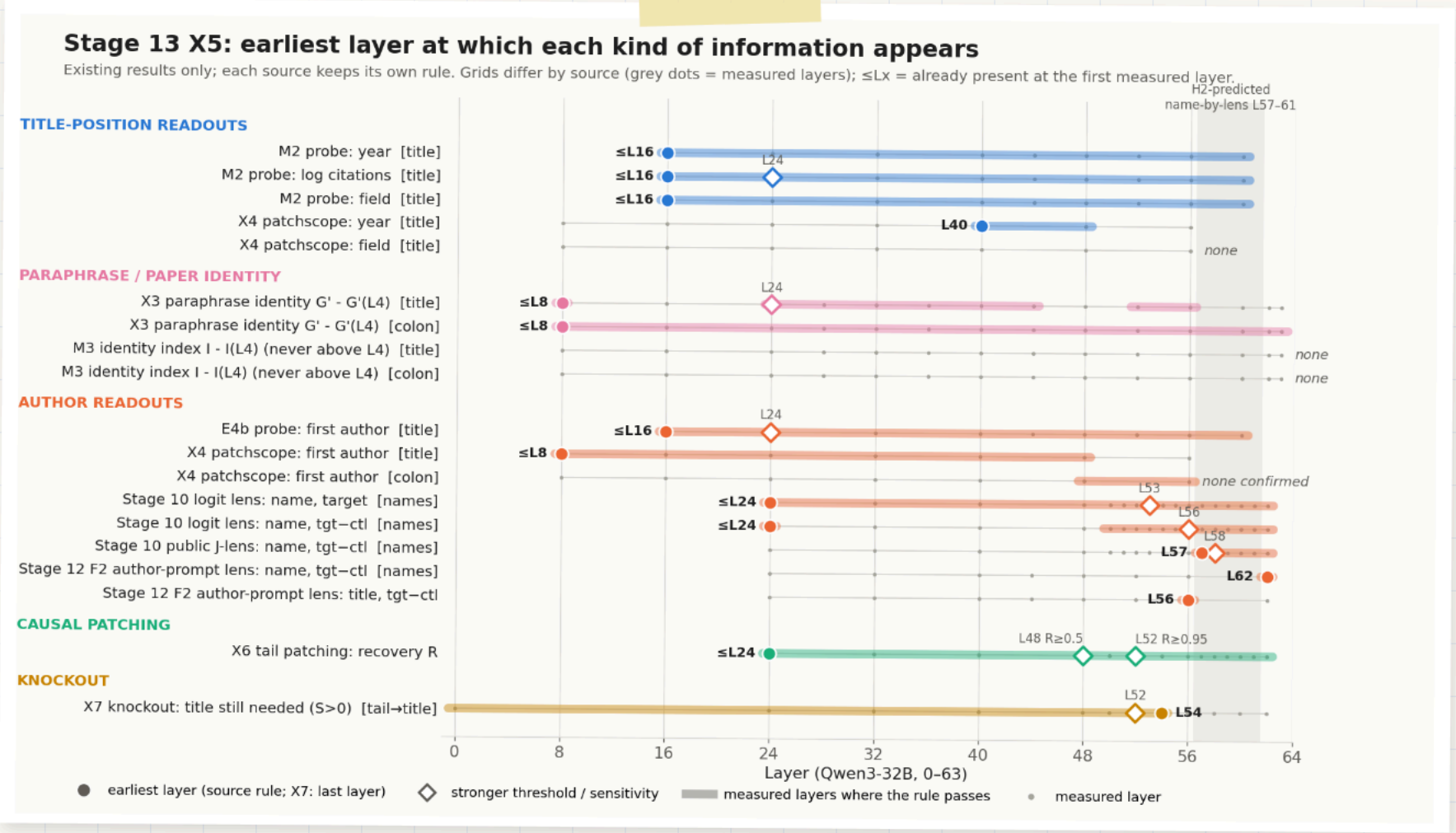
Paper facts are readable at the title early (M2) · H2 (paper identity) partly supported but not established; H3 (hidden author knowledge): partly supported (name-position probe, addendum 11: criterion met for the surname at L40/L48, but lens intervals also exclude 0 and the scales differ, so sizes cannot be compared) · X2 alone: H2 not supported; author subspace can hardly decode authors ($R^2 \leq 0.009$)

怎么算：H3 名字位置 probe spec = probe(true) − probe(false)（Vpd 4,654 篇，团队整群）；H1 question share · 出处：b2_np_report_v2.md、x1_similarity.md

Hypothesis	M1: top word of the lens in middle layers	M2: probe reads non-name attributes	M3: change-of-task control	Current result (report v41, 2026-11-12)
H1 task / format: the middle layers represent "output an author list"	mostly "author", punctuation, common name starts; unrelated to the title	paper attributes not readable or very weak	the same title under different questions separates early in the middle layers	holds at ":" and name positions; at L4–L40, ":" moves 3–5× more when the question changes than when the paper changes (2.2× at L48, 1.3–1.5× at L56–L63)
H2 paper identity: the middle layers represent "which paper this is"	title-related words (copied from the title); change with a false title	field, year (inferred from the title), citations and recognition all readable, and earlier than the names	the same paper under different questions is close in the middle layers and separates by task only late **Result:** not above the L4 text level, not supported	**M3 does not support it; X3 shows a semantic representation beyond wording in the middle layers, but it cannot be separated from paper identity; H2 currently rests on M2 and X1, and M2's Qwen-over-BERT advantage is reference only (addendum 2).** Also, the earliest decodable order in X4 is first author (L8) before year (L40), opposite to the H2 prediction; the X4 recognised − control difference in first-author D (not significant on test) is no longer counted as H2 evidence (text-only BERT baseline +0.088 not weaker than Qwen L8 +0.018). **X2 does not support it.**
H3 author information already there, just not in vocabulary directions	not necessarily	the authors' first token readable by the probe, not by the lens	an author-trained probe transfers across tasks; an author-specific lens (M4) reads names in the middle layers	**Partly supported (2026-11-09, name-position probe, addendum 11):** By the addendum-11 pre-registered criterion, the surname first token meets H3 at L40/L48 (name-position probe, per-donor exclusion; probe L40 +0.069 [0.064, 0.074], L48 +0.103 [0.097, 0.110]; target − control L40 +0.031 [0.019, 0.044], L48 +0.053 [0.038, 0.067]); post-hoc check: the lens intervals also exclude 0 (logit L40 +0.010 [0.005, 0.014]; J-lens L40 +0.048 [0.042, 0.054]) and the two scales differ and cannot be compared in size, so H3 is

Layer order (X5)

Earliest layer at which each kind of information appears · existing results only, each source keeps its own rule



Read early, carried, spoken late

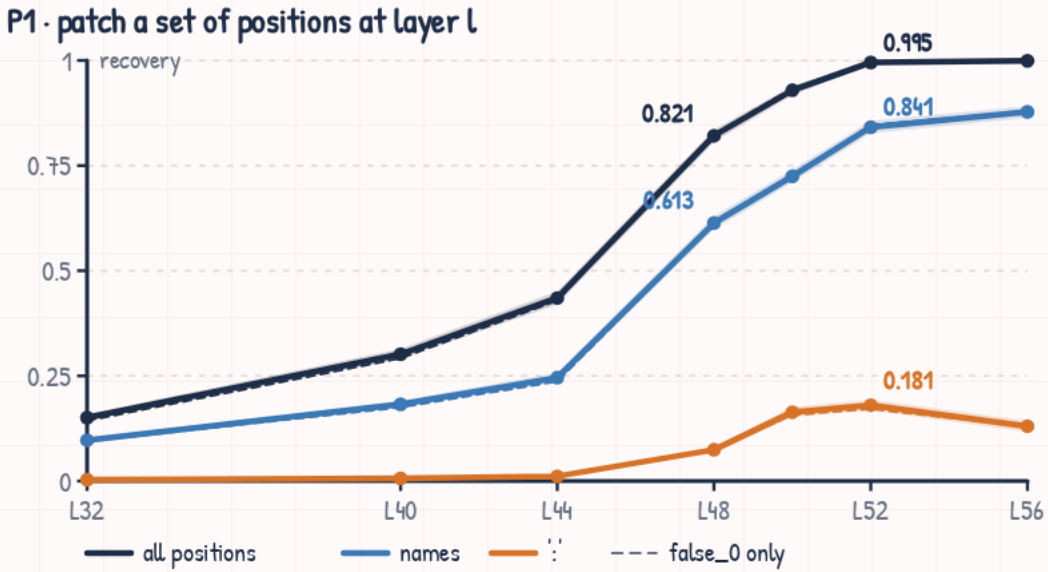
≤L8–L16: title facts and the first-author token

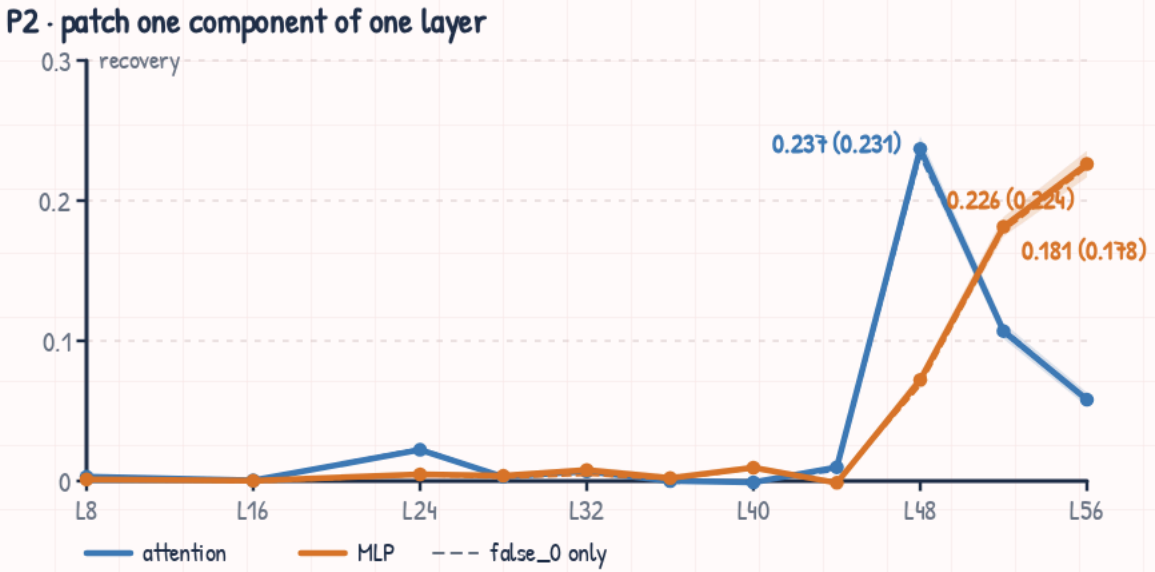
L48–L52: the paper signal reaches the post-title tail (patching R ≥ 0.5 at L48, ≥ 0.95 at L52; knockout costs nothing from L52)

L53–L62: names readable by lens

Stage 12: where the paper information travels

P1 / P2 with five false titles (addendum 10, item 3) · share of the true − false gap recovered at the output · 1,186 papers





怎么算：recovery(l) = mean_p[P̄_p(l) − F̄_p] / mean_p[T̄_p − F̄_p]（P1 位置子集 / P2 单层单组件；5 个假标题平均，细虚线 = 只用 false_0）· Stage 11 样本 1,186 篇 · 团队整群 95% CI · 391_groups/summary_b1_p12f/p12f_mean5.csv（b1_p12f_report.md）

Plotted (mean of 5 false titles [team 95% CI] (false_0)) — all positions after the title: L32 0.151 [0.143, 0.159] (false_0 0.143); L40 0.302 [0.290, 0.314] (false_0 0.292); L44 0.435 [0.420, 0.451] (false_0 0.429); L48 0.821 [0.807, 0.835] (false_0 0.825); L50 0.929 [0.918, 0.940] (false_0 0.933); L52 0.995 [0.990, 1.001] (false_0 0.996); L56 0.999 [0.997, 1.001] (false_0 0.999) | name positions only: L32 0.097 [0.092, 0.103] (false_0 0.094); L40 0.183 [0.174, 0.191] (false_0 0.175); L44 0.246 [0.235, 0.257] (false_0 0.236); L48 0.613 [0.599, 0.630] (false_0 0.612); L50 0.725 [0.710, 0.740] (false_0 0.726); L52 0.841 [0.827, 0.856] (false_0 0.847); L56 0.877 [0.864, 0.891] (false_0 0.878) | ':' only: L32 0.003 [0.002, 0.005] (false_0 0.005); L40 0.006 [0.005, 0.008] (false_0 0.008); L44 0.011 [0.009, 0.013] (false_0 0.011); L48 0.074 [0.068, 0.081] (false_0 0.074); L50 0.164 [0.152, 0.176] (false_0 0.157); L52 0.181 [0.168, 0.193] (false_0 0.172); L56 0.131 [0.117, 0.144] (false_0 0.130) | attention: L8 0.003 [0.002, 0.004] (false_0 0.003); L16 0.000 [−0.000, 0.001] (false_0 0.001); L24 0.022 [0.020, 0.024] (false_0 0.023); L28 0.003 [0.002, 0.004] (false_0 0.003); L32 0.007 [0.006, 0.008] (false_0 0.007); L36 −0.000 [−0.001, 0.001] (false_0 0.001); L40 −0.001 [−0.002, 0.000] (false_0 −0.000); L44 0.010 [0.008, 0.012] (false_0 0.007); L48 0.237 [0.229, 0.246] (false_0 0.231); L52 0.107 [0.101, 0.113] (false_0 0.109); L56 0.058 [0.053, 0.063] (false_0 0.058) | MLP: L8 0.001 [0.000, 0.002] (false_0 0.002); L16 0.000 [−0.001, 0.001] (false_0 −0.001); L24 0.005 [0.003, 0.006] (false_0 0.004); L28 0.004 [0.002, 0.005] (false_0 0.002); L32 0.008 [0.006, 0.010] (false_0 0.004); L36 0.002 [−0.000, 0.004] (false_0 0.001); L40 0.010 [0.008, 0.012] (false_0 0.009); L44 −0.001 [−0.004, 0.001] (false_0 −0.001); L48 0.072 [0.068, 0.077] (false_0 0.068); L52 0.181 [0.174, 0.188] (false_0 0.178); L56 0.226 [0.217, 0.236] (false_0 0.224)

N1 name plausibility: without a title the true list looks "most real" in 0.255 of papers (chance 0.25); with the true title 0.625 [0.615, 0.635]

N2 pure name prior: target − control +0.047 [0.031, 0.062] vs +0.247 with the true title → ≈ 19% of the gap (delete-title prior of §6.12: 0.35)

F2 lens fitted on author prompts: target − control ≤ 0 at L24–L56, +0.238 only at L62

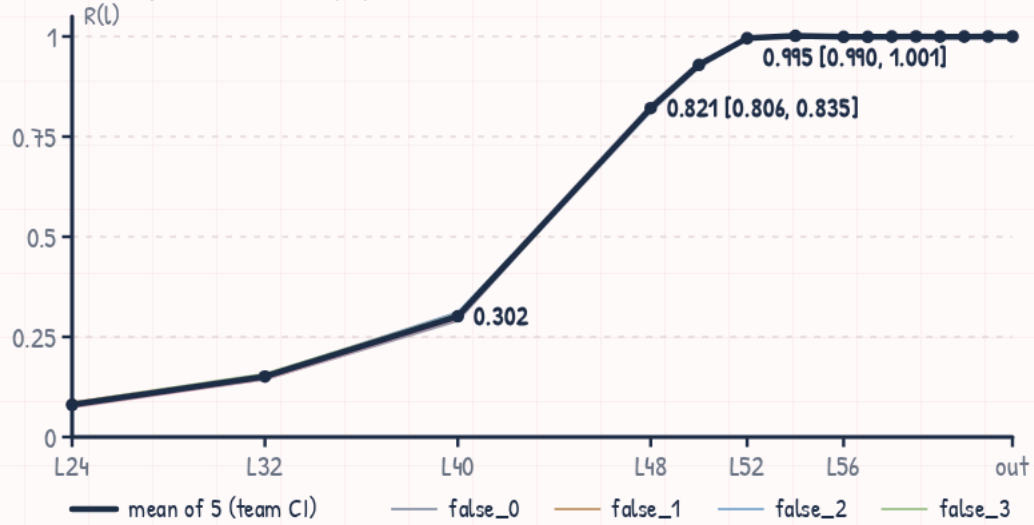
N1 name plausibility: Stage 12 N1, 8,894 R0/R1 papers, paper bootstrap; N1_prior_mc.csv · N2 pure name prior: Stage 12 N2, final L63, 4,621 + 2,220, within bin, paper bootstrap; stage12_report.md · F2 lens fitted on author prompts: Stage 12 F2, name positions, Stage 11 sample, true − 5 false; F2_author_lens.csv

P1 / P2: 391_groups/summary_b1_p12f/p12f_mean5.csv (columns ratio, ratio_lo, ratio_hi, ratio_false0; who = all_authors, group = all); 391_groups/reports/b1_p12f_report.md

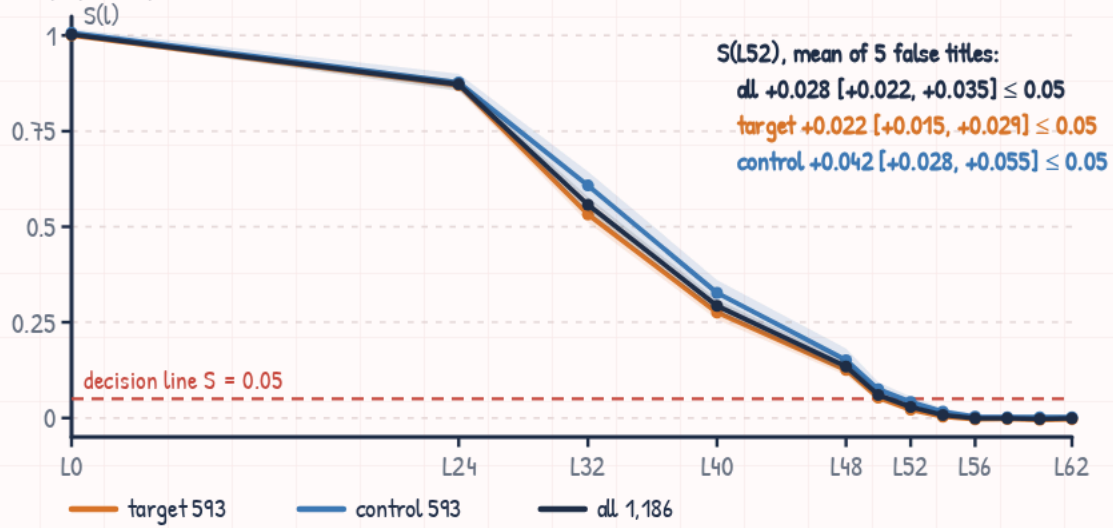
The Stage 11 timing holds for all 5 false titles

X6: replacement with each of the 5 false titles · X7: knockout minus the same knockout on the false-title run (S = T − F)

X6 · recovery R(l), all 1,186 papers



X7 · paper-specific loss S(l), mean of 5 false titles



怎么算：X6：R(l) = mean_papers(nlr_patched@l − nlr_false) / mean_papers(nlr_true − nlr_false)；平均 R = 5 个标题 R 的简单平均（bootstrap 同样逐次平均） · X7：T(l) = mean(kT64 − kT_l) / mean(true − false)；F(l) = mean(kF64 − kF_l) / mean(true − false)；S(l) = T(l) − F(l)（论文特有损失） · 1,186 篇（593 + 593），标题后全部位置，最终输出 · 团队整群 95% CI · summary_x6/x6_recovery.csv；summary_b1_x7f/x7f_knockout.csv

Plotted — mean of 5: L24 0.080 [0.075, 0.086]; L32 0.151 [0.143, 0.159]; L40 0.302 [0.290, 0.315]; L48 0.821 [0.806, 0.835]; L50 0.929 [0.917, 0.941]; L52 0.995 [0.990, 1.001]; L54 1.001 [0.998, 1.004]; L56 0.999 [0.997, 1.001]; L57 0.999 [0.997, 1.001]; L58 0.999 [0.997, 1.001]; L59 1.000 [0.998, 1.002]; L60 0.999 [0.998, 1.001]; L61 1.000 [0.998, 1.001]; L62 1.000 [0.999, 1.001]; L63 1.000 [1.000, 1.000] | false_0: L24 0.077 L32 0.144 L40 0.291 L48 0.825 L50 0.932 L52 0.996 L54 1.001 L56 0.998 L57 0.999 L58 1.000 L59 1.000 L60 0.999 L61 1.000 L62 1.001 L63 1.000 | false_1: L24 0.086 L32 0.156 L40 0.307 L48 0.822 L50 0.926 L52 0.997 L54 1.002 L56 1.000 L57 1.000 L58 1.000 L59 1.000 L60 1.000 L61 1.000 L62 0.999 L63 1.000 | false_2: L24 0.080 L32 0.153 L40 0.310 L48 0.813 L50 0.927 L52 0.996 L54 1.001 L56 1.000 L57 0.998 L58 0.999 L59 1.000 L60 1.000 L61 0.999 L62 1.001 L63 1.000 | false_3: L24 0.085 L32 0.158 L40 0.307 L48 0.823 L50 0.931 L52 0.994 L54 0.999 L56 0.998 L57 0.998 L58 0.998 L59 0.998 L60 0.999 L61 0.998 L62 1.000 L63 1.000 | false_4: L24 0.074 L32 0.145 L40 0.294 L48 0.823 L50 0.929 L52 0.994 L54 1.004 L56 0.999 L57 1.000 L58 1.000 L59 1.001 L60 1.000 L61 1.000 L62 1.000 L63 1.000 | S target: L0 +1.001 [+0.996, +1.005]; L24 +0.870 [+0.856, +0.884]; L32 +0.533 [+0.511, +0.555]; L40 +0.276 [+0.257, +0.294]; L48 +0.126 [+0.112, +0.140]; L50 +0.053 [+0.044, +0.062]; L52 +0.022 [+0.015, +0.029]; L54 +0.003 [−0.001, +0.008]; L56 −0.004 [−0.007, −0.001]; L58 −0.003 [−0.006, +0.000]; L60 −0.005 [−0.008, −0.002]; L62 −0.003 [−0.006, −0.001] | S control: L0 +1.007 [+0.996, +1.018]; L24 +0.877 [+0.854, +0.900]; L32 +0.608 [+0.571, +0.644]; L40 +0.327 [+0.295, +0.361]; L48 +0.151 [+0.122, +0.181]; L50 +0.075 [+0.055, +0.094]; L52 +0.042 [+0.028, +0.055]; L54 +0.016 [+0.008, +0.025]; L56 +0.003 [−0.002, +0.009]; L58 +0.002 [−0.003, +0.008]; L60 +0.002 [−0.003, +0.007]; L62 +0.002 [−0.002, +0.007] | S all: L0 +1.003 [+0.998, +1.008]; L24 +0.872 [+0.861, +0.884]; L32 +0.558 [+0.539, +0.577]; L40 +0.293 [+0.276, +0.310]; L48 +0.134 [+0.120, +0.148]; L50 +0.060 [+0.051, +0.069]; L52 +0.028 [+0.022, +0.035]; L54 +0.008 [+0.004, +0.012]; L56 −0.001 [−0.004, +0.001]; L58 −0.001 [−0.004, +0.002]; L60 −0.003 [−0.005, −0.000]; L62 −0.001 [−0.004, +0.001]

By name-token role: surname first token 0.734 at L48, surname continuation 0.893; all roles ≥ 0.96 at L52

Pre-registered verdict (false_0 only): target S(L52) 0.018 holds, control 0.051 just misses 0.05; 5-title mean: all groups pass

By name-token role: X6 roles, all papers; x6x7_roles.md · Pre-registered verdict (false_0 only): X7 x7_report.md; addendum 10 item 2, b1_x7f_report.md

X6: 381_stage13/summary_x6/x6_recovery.csv (group = all; false_0..4, average) · X7: 381_stage13/summary_b1_x7f/x7f_knockout.csv (title = mean5; S, S_lo, S_hi); x6_report.md, b1_x7f_report.md

Same title, different question

M3: probes across three questions (authors / year / field) · X1: patch A's title state from another question into B's author prompt

- **M3 transfer at the title:** $T \geq 0.92$ at L16–L40, but never above the L4 text level(Stage 13 M3, 9,108 papers, team folds; m3_report.md) (year $T - T(L4)$ −0.03 to −0.06)(same, CIs exclude 0) → M3 does not support H2
- **M3 at the colon:** $T < 0.5$ → the state is already organised by the question (H1)(same; summary_m3/m3_transfer.csv)
- **X1 replacement:** $R \approx 0.996$ (L24), 0.933 (L40), 0.455 (L48) for all authors, whichever question the state came from (old first-author version: 0.99 / 0.92 / 0.42)(Stage 13 X1, test 582 pairs, team CI; x1_report.md); authors − year within ±0.02, except a small +0.019 at L48(same, [0.012, 0.026])
- **Retrieval (added 2026-11-05):** no layer retrieves the paper across questions better than L4(X5 retrieval, 9,108 papers; x5_retrieval.md; post-defence)

Reading

The title state hardly depends on the question, and the replacement effect is task-independent; but nothing goes beyond what the title text alone gives at L4.

Patchscope and paraphrases

X4: decode year / first author / field from the paper's own title state · X3: paraphrased title vs a word-matched other paper

- **X4 first author:** decodable from L8, D = +0.046 [0.035, 0.059](Stage 13 X4, test split, team CI; x4_report.md) (all authors also from L8: +0.048 [0.039, 0.058]); year only from L40, +0.013 [0.008, 0.018](same); field never → order opposite to the H2 prediction
- **But:** the L8 signal sits on the surname (+0.066), where the text-baseline check failed(X4 role split, 1,178 papers; x4_rolesplit.csv) → may be explained by the title text
- **X3 paraphrase:** closer to the original than a word-matched paper beyond L4, peak L36 +0.16(Stage 13 X3, test 552 papers, G′ − G′(L4), team CI [0.146, 0.171]; x3_report.md); behaviour keeps ≈ 90% of the author preference(all authors, all tokens 0.897 [0.868, 0.926]; old first-author version 0.90 [0.86, 0.94])
- **Post-hoc checks:** a sentence-embedding model without author knowledge shows a comparable advantage, and target = control(X3 text baselines and group split; x3_textbase_recovery.md, x3_groupsplit.md) → meaning, not separable from paper identity

H2 / H3 so far

H2 partly supported but not established (rests on M2 and X1; X4 recognition gap no longer counted); H3: partly supported by the name-position probe (addendum 11); X4 alone cannot exclude the title text.

X2: swap only a subspace

Patch only the paper-attribute (S_id, 9-dim) or the author (S_auth, 10-dim) component of A's title states into B · test 582 pairs · team-cluster 95% CI

- **Carry:** S_id at L40 moves +0.005 of the gap (random +0.003, full replacement +0.920); S_auth at L48 moves +0.018 (random +0.003, full +0.428)(Stage 13 X2, test 582 pairs; x2_report.md Verdict)
- **Pre-registered verdict:** H2 not supported (R_id 0.005 < 0.5 × R_full 0.460); H3 passes (R_auth − R_id +0.014 [+0.008, +0.020])(same, L40 / L48)
- **But:** the author subspace hardly decodes authors (held-out $R^2 \leq 0.003$; first-author token hit rate 0.078–0.079, below the constant baseline 0.080) and moves only about 4% of the full replacement(subspace fits on 5,041 training papers, 5 team folds; x2_report.md)
- **Remove arm:** setting either component to the mean costs no more than a random subspace (+0.001)(test, L_S − L_rand; same)

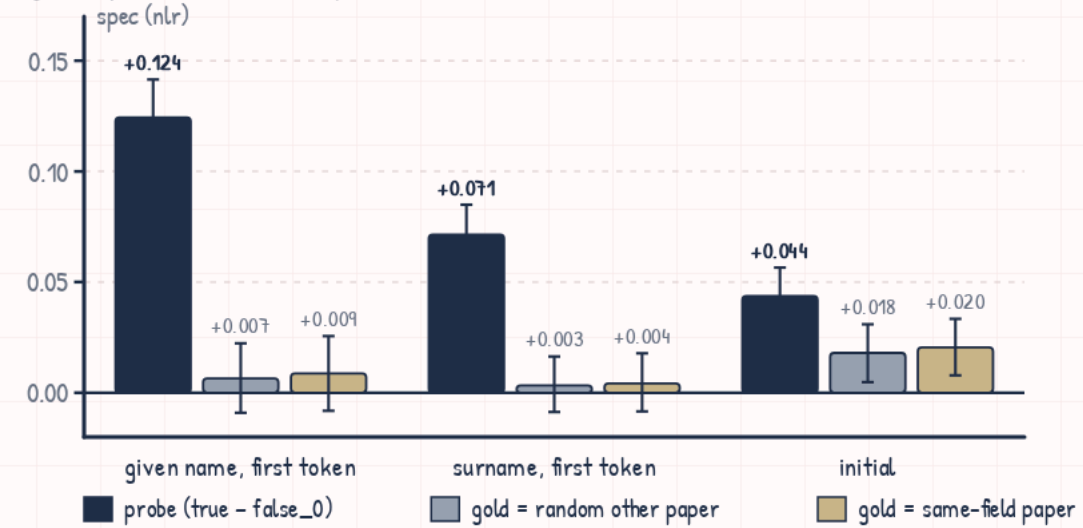
Reading

H3 passes by the pre-registered rule and H2 is not supported, but the author subspace cannot decode authors, so this is no strong evidence for H3. The information carried from the title to the names lies almost entirely outside both subspaces.

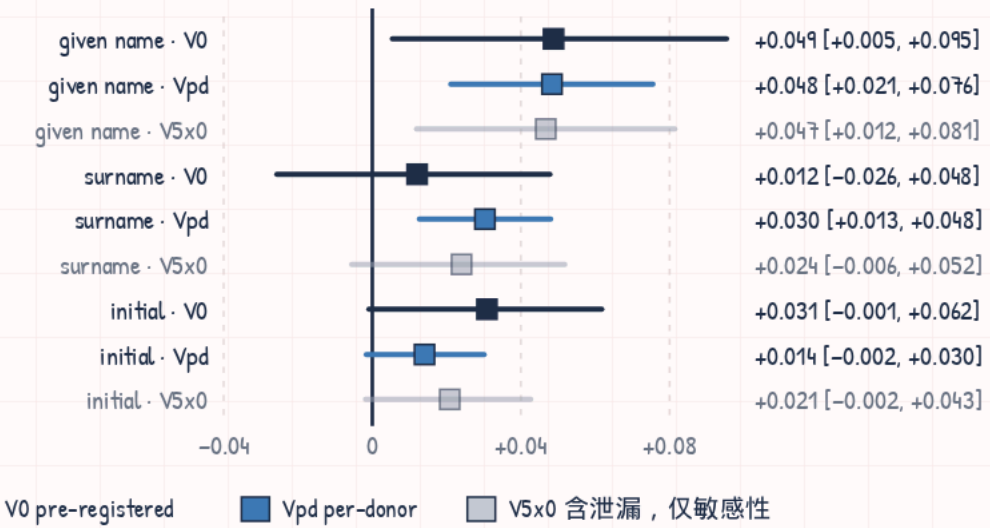
The probe on every name token

Title-position probe, all authors, all name tokens, split by role · L40 · team-cluster 95% CI

Signal by role at L40, with permutation controls



Target − control at L40 (within citation bin), three versions



怎么算：spec(p) = score(p, X_true) − score(p, X_false_0)，每个名字 token 按角色取名次 → nlr；T − C：target−control = Σ_b w_b [mean_{target,b} − mean_{control,b}]，w_b = n_t n_c/(n_t+n_c) 归一化 · L40，名 1,706 / 姓 1,395 / 首字母 1,590 篇（V0）；Vpd 3,046 / 4,654 / 4,812 篇 · 团队整群 95% CI · alltok_controls.csv；perdonor_by_run_layer.csv

Plotted — given name, first token · probe (true − false_0): +0.124 [+0.108, +0.142] (n 1,706) | given name, first token · gold = random other paper: +0.007 [−0.009, +0.022] (n 1,706) | given name, first token · gold = same-field paper: +0.009 [−0.008, +0.026] (n 1,706) | surname, first token · probe (true − false_0): +0.071 [+0.058, +0.085] (n 1,395) | surname, first token · gold = random other paper: +0.003 [−0.009, +0.016] (n 1,395) | surname, first token · gold = same-field paper: +0.004 [−0.008, +0.018] (n 1,395) | initial · probe (true − false_0): +0.044 [+0.030, +0.057] (n 1,590) | initial · gold = random other paper: +0.018 [+0.005, +0.031] (n 1,590) | initial · gold = same-field paper: +0.020 [+0.008, +0.033] (n 1,590) | T − C given name, first token V0: +0.049 [+0.005, +0.095] (n 1,706) | T − C given name, first token Vpd: +0.048 [+0.021, +0.076] (n 3,046) | T − C given name, first token V5x0: +0.047 [+0.012, +0.081] (n 1,706) | T − C surname, first token V0: +0.012 [−0.026, +0.048] (n 1,395) | T − C surname, first token Vpd: +0.030 [+0.013, +0.048] (n 4,654) | T − C surname, first token V5x0: +0.024 [−0.006, +0.052] (n 1,395) | T − C initial V0: +0.031 [−0.001, +0.062] (n 1,590) | T − C initial Vpd: +0.014 [−0.002, +0.030] (n 4,812) | T − C initial V5x0: +0.021 [−0.002, +0.043] (n 1,590)

Surname first token above the initial: +0.071 vs +0.044, difference +0.028 [+0.011, +0.045] → both pre-registered checks pass; E4b (first token only, mostly initials) underestimated it

Permutation controls: ≈ 0 on surname and given-name tokens; ≈ +0.02 on initials → about half of the initial's signal is not paper-specific

Target − control: given name positive in all three versions; surname not significant pre-registered, positive only post hoc (per-donor) → surname evidence weak

Surname first token above the initial: e4b_alltok.md §5 · Permutation controls: e4b_alltok.md §7 · Target − control: b1_e4b5_report.md

Bars: 381_stage13/summary_alltok/alltok_controls.csv (layer 40) · forest: 381_stage13/summary_b1_e4b5/perdonor_by_run_layer.csv (tc, tc_lo, tc_hi; V0 / Vpd / V5x0) · e4b_alltok.md, b1_e4b5_report.md

Name-position probe: H3 partly supported

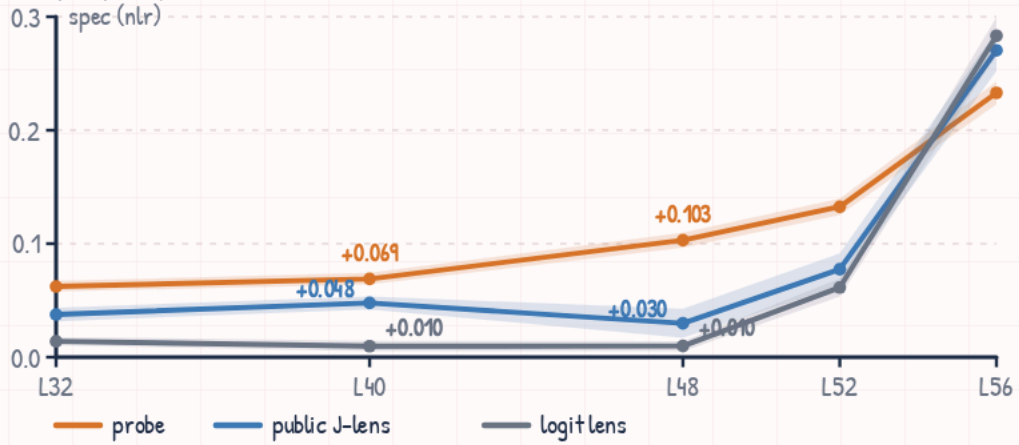
Addendum 11 A · per-donor version Vpd (D11-1) · surname first token · all papers 4,654 · team-cluster 95% CI

1 title in prompt → 2 residual at name-predicting position, layer l → 3 ridge probe → 4 → model's own L63 state → 5 decode among 9,137 names → nlr

spec = probe(true) − probe(false)
false = mean of the usable false titles (Vpd)

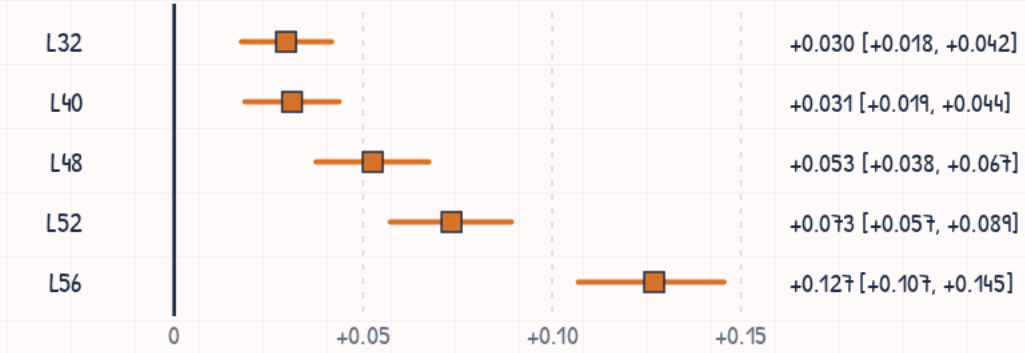
按补充 11 预注册判据通过（姓首 token L40/L48）；事后检验：lens 区间同样不含 0、尺度不同无法比较大小，因此降为部分支持

spec by layer: probe vs lenses



different rank scales (9,137 names vs full vocabulary): compare sign and CI, not size

Probe target − control by layer (within citation bin)



Initials (L40): permutation-control row +0.020 [+0.015, +0.026] ≈ half of the signal +0.045 [+0.039, +0.051]
首字母的对照行约为信号的一半：约一半信号不是论文特有的

怎么算：名字位置 probe：ridge 从第 l 层"预测名字 token 的位置"的残差拟合到同一位置模型自己的 L63 状态，在 9,137 个名字 token 内求名次 → nlr；spec = probe(true) − probe(false)（方法手册 定义：名字位置 probe spec） · T − C：档内合并差 · Vpd 4,654 篇（姓首 token），层 L32–L56 · 团队整群 95% CI · summary_b2_np/np_lens_compare_v2.csv；np_role_layer_v2.csv

Plotted — probe (9,137 name tokens): L32 +0.062 [+0.058, +0.067]; L40 +0.069 [+0.064, +0.074]; L48 +0.103 [+0.097, +0.110]; L52 +0.133 [+0.125, +0.140]; L56 +0.233 [+0.223, +0.243] | public J-lens (full vocabulary): L32 +0.038 [+0.032, +0.044]; L40 +0.048 [+0.042, +0.054]; L48 +0.030 [+0.017, +0.043]; L52 +0.078 [+0.063, +0.092]; L56 +0.270 [+0.252, +0.288] | logit lens (full vocabulary): L32 +0.014 [+0.011, +0.018]; L40 +0.010 [+0.005, +0.014]; L48 +0.010 [+0.005, +0.014]; L52 +0.061 [+0.054, +0.069]; L56 +0.283 [+0.267, +0.299] | probe T − C L32: +0.030 [+0.018, +0.042] | probe T − C L40: +0.031 [+0.019, +0.044] | probe T − C L48: +0.053 [+0.038, +0.067] | probe T − C L52: +0.073 [+0.057, +0.089] | probe T − C L56: +0.127 [+0.107, +0.145] | Initials (L40): permutation-control row +0.020 [+0.015, +0.026] ≈ half of the signal +0.045 [+0.039, +0.051]

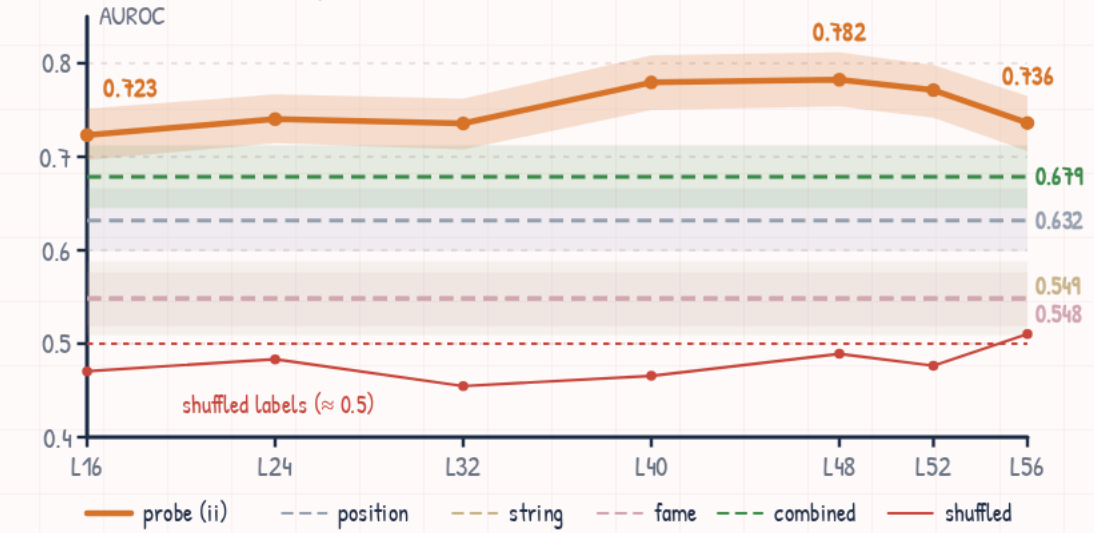
Names the model writes itself: true vs fabricated

C24 · addendum 15, item 2-11b · 518 target papers, 671 true names (M) vs fabricated (H) · team-cluster 95% CI

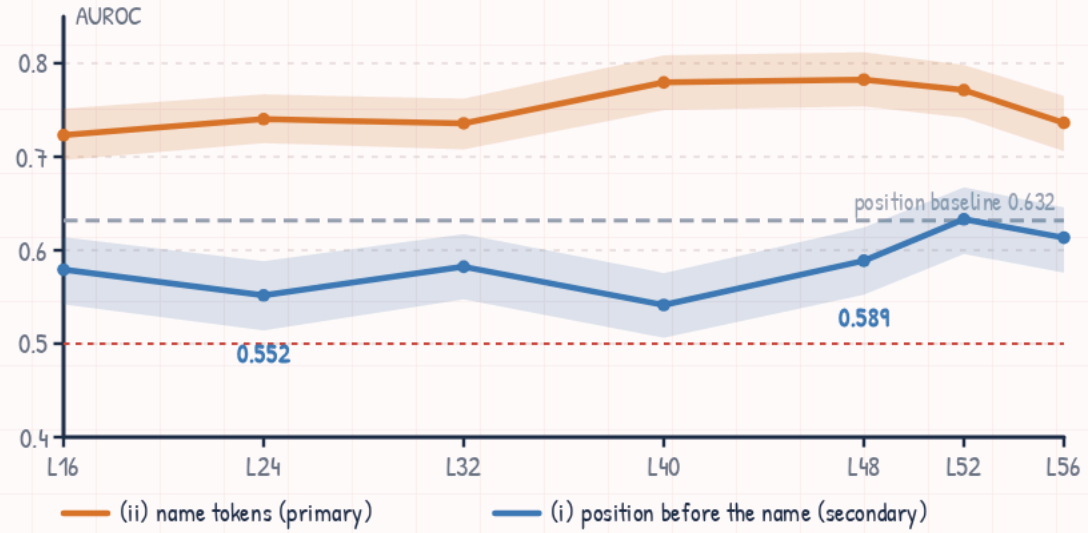
初步：输出置信度基线暂停

1. model writes the author list itself
2. residuals saved during generation
3. label each written name true / fabricated
4. within-paper AUROC

(ii) on the name tokens: probe vs baselines



Before the name (i) vs on the name (ii)



怎么算：论文内 AUROC：每篇 target 论文（≥ 1 个真作者 M 且 ≥ 1 个编造名字 H）内的全部 M-H 名字对，probe 分数 M > H 记 1、并列记 0.5，取平均（方法手册 定义：2-11b 论文内 AUROC） · 518 篇 target 论文、3,388 个 M-H 对，L16–L56 · 团队整群 95% CI · b2_gennames_b/summary/probe_within.csv；robust_posthoc.csv

Plotted — position: 0.632 [0.599, 0.666] | name / string features: 0.549 [0.509, 0.588] | fame (post hoc): 0.548 [0.519, 0.576] | combined (post hoc): 0.679 [0.645, 0.712] | shuffled: L16 0.470 [0.434, 0.506]; L24 0.483 [0.449, 0.516]; L32 0.455 [0.420, 0.490]; L40 0.466 [0.434, 0.496]; L48 0.489 [0.456, 0.525]; L52 0.477 [0.443, 0.513]; L56 0.510 [0.478, 0.543] | probe (ii) name tokens: L16 0.723 [0.696, 0.751]; L24 0.740 [0.715, 0.767]; L32 0.736 [0.708, 0.762]; L40 0.780 [0.750, 0.808]; L48 0.782 [0.754, 0.812]; L52 0.771 [0.742, 0.798]; L56 0.736 [0.706, 0.765] | probe (i) pre-name position: L16 0.579 [0.542, 0.614]; L24 0.552 [0.514, 0.588]; L32 0.582 [0.548, 0.617]; L40 0.541 [0.506, 0.576]; L48 0.589 [0.552, 0.624]; L52 0.633 [0.596, 0.667]; L56 0.613 [0.576, 0.646]

补充 15 预注册判据：(ii) 在 L16–L56 七层都同时高于排位基线和名字特征基线（例如 L48 0.782 [0.754, 0.812]）；(i) 名字写出之前读不出（不高于排位基线）。

前提：chat 模板下现场生成（不是 teacher forcing）；标签来自规则匹配；control 只作描述；与 H3 的关系未预注册；输出置信度基线（生成时 log-prob / 熵）暂停，待定。

381_stage13/b2_gennames_b/summary/probe_within.csv (auroc_within, lo, hi; position / string baselines) · robust_posthoc.csv (fame, combined, shuffled CI) · b2_gennames_b_report.md, b2_gennames_b_robust_posthoc.md

What we learned: read early, spoken late

Read early	Causally the title is read in L24–L50 and gathered at the name positions by L52, then no longer needed. A probe at the title token reads a little of it from L24 (re-checked).
Spoken late	At the name positions, paper-specific author knowledge becomes lens-readable only from ~L50, fastest over L57–L61.
No lens lead	Read with a lens at the names, target is not ahead of control in middle layers; the +0.21 gap is at the end (Stage 10, final L63, §6.4) (probe at the title: target − control +0.03–0.08, re-checked (Stage 13 E4b, first name token only)).
Real knowledge	Name prior is ~1/3 of the gap overall but ~0 on surnames (Stage 12 A, 0.087 / 0.247, §6.12); the rest is paper-specific and grows with citations.
Lenses agree	Jacobian lenses read more in middle layers, but nothing target-specific.
Name format	Initials carry little; full given names carry as much as surnames and help predict them.

Limits

Done: Stage 12

- Single-position patching (P1); attention vs MLP (P2, hook check passed)
- Name priors (N1, N2); author-prompt lens (F1, F2): middle layers still not readable

Done: Stage 13

- Probes read year, citations, field at the title from ≤L16 (M2); identity not above the L4 text level (M3)
- Replacement is task-independent (X1); paraphrase meaning beyond wording in L28–L40, not separable from paper identity (X3)
- First author decodable at L8, may be text-explainable (X4); 5 false titles confirm the Stage 11 timing (X6, X7)

Limits

- One model (Qwen3-32B)
- Teacher-forced preference, not generation
- Labels from the team lead's package; labelling code (elm) not available
- H2 (paper identity) partly supported but not established; H3 (hidden author knowledge): insufficient evidence

Added after the defence

Added after the defence (2026-11-05)

Group comparisons (plan fixed before the results: addenda 7, 8) · CPU leftovers (post-hoc) · experiments still running

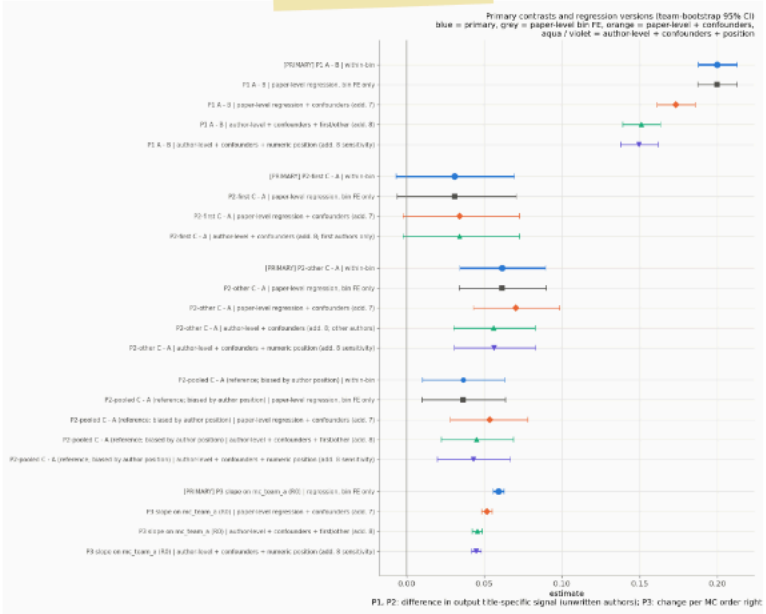
1. Group comparisons A / B / C / D: three findings, intervention re-runs (E1 and P1 / P2 numerator / denominator), recognition
2. CPU leftovers on existing data: X5 retrieval curve, M3 scores, N2 by layer, paper vs team intervals, M5
3. Redo programme (addenda 10–14): done, batch 1 (complete), batches 2 and 3 (in progress); limits that cannot be redone

Added after the defence · groups

Group comparisons: three findings

Plan fixed before the results (addenda 7, 8) · unwritten authors only · output (L63) title-specific signal · within citation bin · team-bootstrap 95% CI · Holm over 4 tests

- **1 Recognition:** with no author written (R0), recognised papers (A) beat unrecognised ones (B) by **+0.200 [0.188, 0.213]**(P1 A − B, 3,417 / 2,148 papers, Holm p ≤ 0.002; primary.csv); with confounders +0.173 [0.161, 0.186](paper level, 5,565), author level + position +0.151 [0.139, 0.163](19,893 authors)
- **2 Graded:** each extra multiple-choice order right adds **+0.059 [0.056, 0.063]**(P3, 8,288 R0 papers, bin FE; Holm p ≤ 0.002) (confounders +0.046 [0.042, 0.049](author level)); ≈ 0.17 with none right, ≈ 0.46 with all four(R0, 927 / 4,105 papers; stage10_curves.csv)
- **3 Partial recall helps little (≈ ⅓ of recognition), output only:** C − A co-authors **+0.061 [0.034, 0.090]**(P2-other, 390 / 2,826 papers, Holm p ≤ 0.002), first author +0.031 [−0.006, 0.069], n.s.(P2-first, 292 / 3,199); ≈ 0 at L48–L62 with the logit lens, −0.07 to −0.11 at L48–L59 with the public J-lens(stage10_contrasts.csv, unwritten_other). The earlier ≈ +0.11 used all authors (circular)(r0_r1_analysis.md §2)
- **Reference only:** R2 written authors +0.844(200 papers; summary_table.md); only 28 R2 papers have unwritten authors(same)



Primary contrasts (blue) and regression versions; team-bootstrap 95% CI (391_groups/figures/fig3)

Added after the defence · re-runs

Re-runs: numerator and denominator

Addendum 8 · E1 recovery R = mean(patched − false) ÷ mean(true − false) · unwritten authors · team-cluster 95% CI

group (unwritten authors)	papers	denominator: true − false	E1 numerator L48	E1 R(L48)	E1 R(L52)
A R0 × recognised	557	+0.407 [0.386, 0.430]	+0.330	+0.811 [0.790, 0.832]	+0.988 [0.980, 0.996]
B R0 × not recognised	586	+0.214 [0.196, 0.234]	+0.186	+0.869 [0.835, 0.906]	+1.016 [1.002, 1.031]
U0 R0 × uncertain	599	+0.293 [0.271, 0.316]	+0.248	+0.844 [0.814, 0.877]	+1.000 [0.988, 1.012]
C R1 × recognised (addendum 11 B)	462	+0.559 [0.529, 0.590]	+0.408	+0.731 [0.705, 0.757]	+0.985 [0.977, 0.994]
R2 written authors (reference)	200	+0.845 [0.798, 0.897]	+0.566	+0.670 [0.630, 0.707]	+0.948 [0.927, 0.965]

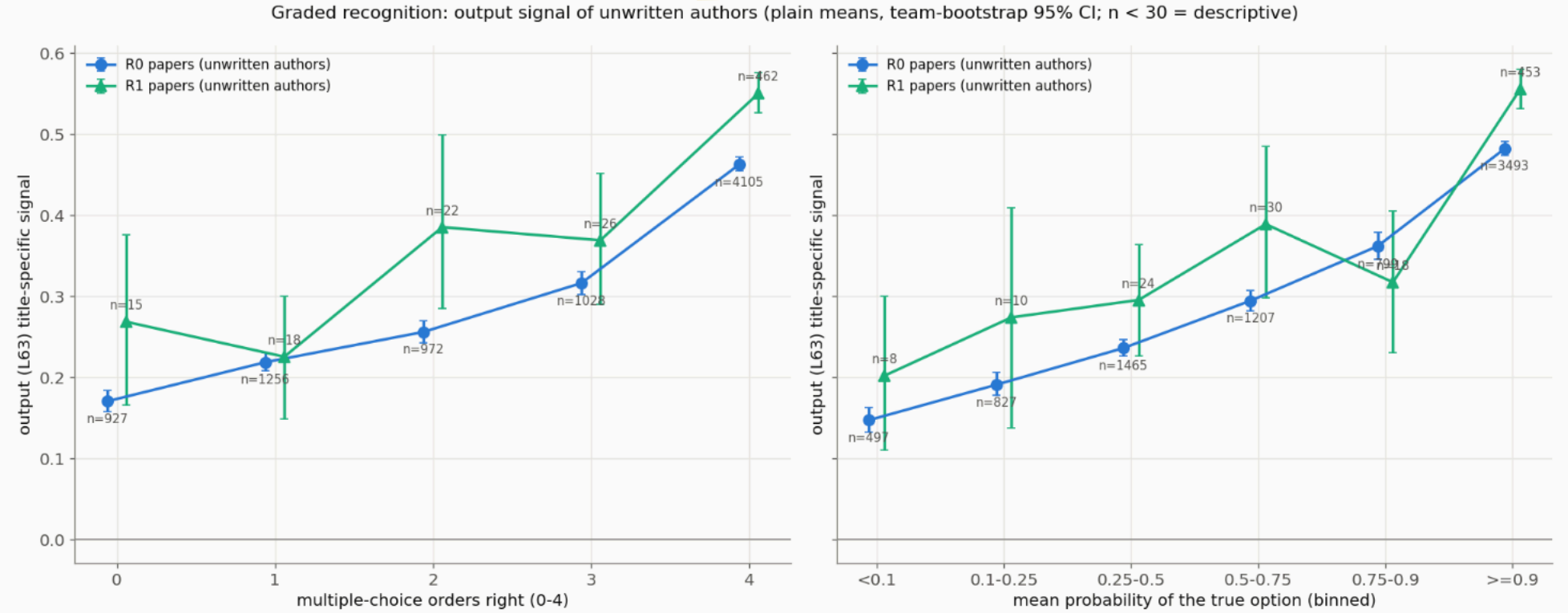
B's higher ratio is a small-denominator effect: numerator B − A −0.010, −0.040, −0.144, −0.184 at L24 / L40 / L48 / L52, all CIs below 0(e1_e2_num_den_B_minus_A.csv) — B carries no more absolute paper-specific signal than A at any layer.

U0 timing as A and B (descriptive); R2 upper reference only; C completed to 462 papers: within bin (258 of 462 C papers, 6 bins) same timing as A, larger gap (C − A +0.075 [+0.024, +0.130]); D 33 / R2-unwritten 28 papers → descriptive(rerun_report.md Counts; b2_cd_report.md)

Added after the defence · groups

Recognition changes the size, not the timing

Unwritten authors, output (L63) signal, plain means with team-bootstrap 95% CI; n < 30 descriptive



Left: multiple-choice orders right (0–4); right: mean probability of the true option. Blue R0, green R1 (391_groups/figures/fig2)

Re-runs: over 0–4 orders right the denominator rises 0.185 → 0.407, while E1 R(L48) stays flat, 0.90 → 0.81 (R0 papers A + U0 + B, 241 / 557 papers at 0 / 4; rerun_report.md "Continuous recognition")

怎么算： 未写出作者的 spec（L63，真 − 5 个假标题平均）按识别程度分组的普通均值 · R0 927–4,105 / R1 15–462 篇 · 团队整群 · 出处：stage10_curves.csv、rerun_report.md

Added after the defence · re-runs

P1 / P2: the same small-denominator effect

Stage 12 single positions (P1) and attention / MLP (P2) split by group · unwritten authors · paired team bootstrap 95% CI

patched part	layer	numerator B − A	ratio B − A	numerator B/A
P1 all name positions	L48	−0.105 [−0.123, −0.087]	+0.052 [+0.011, +0.098]	0.57
P1 all name positions	L56	−0.183 [−0.209, −0.157]	−0.050 [−0.087, −0.012]	0.50
P2 attention	L48	−0.029 [−0.038, −0.021]	+0.057 [+0.030, +0.086]	0.66
P2 MLP	L56	−0.041 [−0.053, −0.029]	+0.008 [−0.030, +0.044]	0.54
P1 ':' only	L48	−0.000 [−0.007, +0.005]	+0.050 [+0.026, +0.071]	0.99
P1 ':' only	L52	−0.013 [−0.024, −0.002]	+0.073 [+0.038, +0.106]	0.79

B's higher P1 / P2 ratios also come from the small denominator: B's numerator is below A's at the name positions, in attention and in the MLP; at the ':' token the numerators are similar in L44–L48 (B/A 0.72–0.87 with five false titles; false_0 0.78–0.99), and B's ratio is higher only because its denominator is about half of A's(A 557 / B 586 papers; denominator B − A −0.192, B/A 0.53; p1_p2_num_den_B_minus_A.csv; e1_num_den.md P1/P2)

Added after the defence · CPU

CPU leftovers on existing data (post-hoc)

No new GPU runs · descriptive only · not in the pre-registration except the X5 retrieval curve

- **X5 retrieval curve (pre-registered, computed now):** top-1 retrieval of the same paper across questions is never above L4 (M3 cache, 9,108 papers; x5_retrieval.md) (title token 1.000 up to L44, 0.734 at the output) (same, team CI)
- **M3 behaviour scores:** year target − control +0.007 [0.003, 0.012]; all authors, all name tokens +0.234 [0.215, 0.252] (2-10; old first-author first-token version +0.147 [0.096, 0.191]) (output, 4,621 / 2,220, within bin, team CI; m3_scores.md)
- **N2 by layer:** pure name prior target − control +0.047 at the output, +0.039 at logit L62; public J-lens L48 −0.133 (true title) (Stage 12 N2 rescored by layer; n2_layers.md) → only L62 reproduces the output pattern
- **Paper vs team intervals:** width ratio 0.97–1.00 (median); "excludes 0" changes in 2 of 1,155 rows (X1, X4, X6, X7, M3; v42_paper_vs_team.md)
- **M5 true vs false title at the name positions:** output 0.919; logit L24 0.516 → L62 0.899 (share of pairs, all papers × 5 false titles, team CI; m5_report.md); probe at title / colon AUROC 0.50–0.52 (same, Part B)

Added after the defence · redo programme

Redo programme (addenda 10–14)

All rules fixed before the results · reproducing the published numbers is a hard check for every item · status 2026-11-08

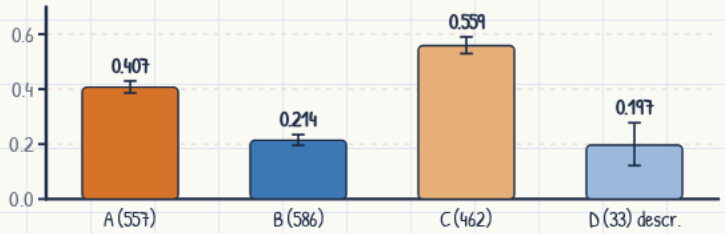
- **Done:** X7 with five false titles; all-token probe (addendum 9); CPU items C-1 to C-5; C-3b (no-exclusion sensitivity, includes leakage, not a conclusion)
- **Batch 1, done:** X1–X4 for all authors → conclusions unchanged (X1 task-independent, small L48 author – year +0.024; X3 keeps ≈ 90%; X4 from L8; X2 moves 0.032, subspace $R^2 \leq 0.009$, X2 alone leaves H3 insufficient)(b1_xall_report.md) · E4b, per-donor exclusion (4,654 papers, ~1.5 false titles each): surname L40 +0.076 [0.068, 0.083], surname – initial +0.033(b1_e4b5_report.md)
- **Batch 1, done:** P1 / P2 with five false titles (five-title mean primary, false_0 beside): every mean5 – false_0 CI includes 0 → conclusions unchanged; colon numerator B/A 0.72–0.87; p9: 694–711 papers per false title, different sets, unweighted(b1_p12f_report.md)
- **Batch 2 (addenda 11, 12):** running; C / D completion done (C 462 papers; within bin 258 of 462 vs A: same timing); 2-11 stopped by its hard check → 2-11b (addendum 15): done: pre-registered verdict passes at all 7 layers (C24); 2-2, 2-3, 2-4, 2-5 (name-position probe: H3 partly supported), 2-8, 2-10 done; Patchscope (2-7, 2-9) done (descriptive only); 2-1: deviation D12-5 (gate reference computed in the wrong precision, f64 vs the fp32 log; not a relaxed criterion); gate fixed to fp32 vs fp32 and resumed, gate v4 queued; deviations recorded (D12-1 to D12-3; 2-7 source = year-free X4 prompt)(prereg_addendum12_deviations.json)
- **Batch 3 (addendum 14):** 3-1 Stage 10 L41–L47, L49: done · 3-2 L0–L23: running (194/504) · 3-3 Stage 7.1B fast path: stopped (deviation D14-1; equivalence with the old per-prefix results not confirmed); 3-4 / 3-5 (Stage 8 GPU W) not run(prereg_addendum14.json)
- **Not done:** Stage 8 GPU re-run of the original W (user decision 2026-11-06; CPU proxy C-2 instead)(c13_C2.md)

Added after the defence · limits

Limits that cannot be redone

Written 2026-11-06; last three items added 2026-11-08 (v30)

- **One model:** the groups (recognised or not, authors written or not) come from Qwen3-32B's own labels; another model would need new labels
- **Teacher forcing:** every score is the rank the model gives the correct name once it is in the prompt — a preference, not generation (C24 使用模型自生成文本，非 teacher forcing)
- **Label pipeline:** the label package author's elm code and run.json are not on the cluster, so generation length, thinking mode and the multiple-choice setup cannot be re-checked
- **High-citation bins:** the 4097+ bin has no B / D papers(84 target, 0 control; m3_scores.md §2), so within-bin comparisons cannot use it
- **Layers and samples:** Stage 10 samples L24–L40 every 8th layer (no L25–L31, L33–L39; no plan to fill; L41–L47, L49 filled by 3-1); interventions use 1,186 of 9,108 papers; denominator effects (C 0.559 vs A 0.407)
- **Post hoc and single title:** parity tolerance 0.25 → 0.5 was used, a post-hoc change (main round: 10 of 25 pairs above 0.25, max 0.367; size matches bf16 rounding, up to ~0.46); M5 part B and the E4b X1-controls pipeline use false_0 only, no plan
- **Layer gap, stopped item:** X4 year onset somewhere in L33–L40; Jacobian vs logit jump between L32 and L40; X4 first-author peak could be L33–L47 · 2-11 stopped (first-name reproduction 0.880 < 0.90)



Denominator effect (L03): E1 true − false gap by group, unwritten authors

怎么算：分母 = mean_p[T̄_p − F̄_p]（recovery(I) = mean_p[P̄_p(I) − F̄_p] / mean_p[T̄_p − F̄_p] 的分母，false_0）· 未写出的作者，Stage 11 样本 + 补充 11 B 补全（C 462、D 33 篇只作描述）· 团队整群 95% CI · 出处：391_groups/b2_cd/summary/cd_group_values.csv（b2_cd_report.md）

Plotted — A: 0.407 [0.386, 0.430] (n 557) | B: 0.214 [0.196, 0.234] (n 586) | C: 0.559 [0.529, 0.590] (n 462) | D: 0.197 [0.122, 0.277] (n 33)